

Special Participation E

From Handwritten Notes to Exam-Oriented Review (Gemini)

I built an AI tool that reads my handwritten lecture notes and generates structured review priorities, and I analyze how well it distinguishes core concepts from superficial coverage.

1. Motivation & Problem Setting

Many students take notes in diverse formats, including handwritten notes on paper or tablets. Unlike lecture slides or PDFs, handwritten notes are often incomplete, unstructured, and highly subjective.

In this project, I explore whether a modern multimodal LLM (Gemini) can act as an AI-enhanced learning tool that (1) reconstructs structured study notes from handwritten lecture notes provided as images and (2) identifies which parts of those notes are most critical for exam preparation rather than merely frequent or visually emphasized.

2. AI Tool Design

2.1 Input Modality

The input to the system is a photo of handwritten lecture notes. This choice is intentional: handwritten notes better reflect how students actually capture information in lectures, including omissions, shorthand, and emphasis markers that are not present in official materials.

2.2 Prompt Design

*The following image contains my handwritten lecture notes.
They are messy, incomplete, and reflect my personal emphasis during the lecture.*

Your tasks are:

- 1. Reconstruct a clean, structured version of the notes with clear sections and logical flow.*
- 2. Identify concepts that are most important for exam review.*
- 3. For each identified concept, briefly explain why it is exam-critical.*
- 4. Explicitly flag any content that appears emphasized in the notes but is likely not central for assessment.*

Do NOT merely rewrite the notes verbatim.

Focus on structure and prioritization rather than completeness.

3. Interaction Trace

Machine Learning Optimization & Regularization

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Analysis

This is a clean, structured reconstruction of your lecture notes, focusing on the core concepts for effective exam review.

Optimization and Regularization in Machine Learning

Helpful for understanding: Clearly states the high-level goal and the formulas.

1. Model and Optimization

The fundamental goal is to find the optimal model parameters $\mathbf{w}$ that map an input $\mathbf{x}$ to an output $f_{\mathbf{w}}(\mathbf{x})$ based on observed $(\mathbf{x}, y)$ data.

Concept	Definition/Formula	Exam-Critical Rationale
Empirical Risk Minimization (ERM)	$R(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n L(y_i, f_{\mathbf{w}}(\mathbf{x}_i))$	Core principle of finding the parameter $\mathbf{w}$ that minimizes the average loss on the training data. This is the starting point for most learning algorithms.
Key Assumption	Data $(\mathbf{x}, y)$ are drawn from the true, underlying distribution $P(\mathbf{x}, y)$.	Essential for understanding why ERM is used—to approximate the true expected risk $E_{(\mathbf{x}, y) \sim P(\mathbf{x}, y)} [L(y, f_{\mathbf{w}}(\mathbf{x}))]$.

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2. Optimization Challenges and Solutions

ERM faces several challenges, primarily addressed by using a **validation set** and **regularization**.

Challenge	Description	Solution/Technique	Exam-Critical Rationale
Challenge 1: Generalization	The model performs well on the training data but poorly on unseen test data (overfitting). This is often due to the model being too complex relative to the data.	Use a hold-out test set (or cross-validation) to estimate the true generalization error.	Testing the model's ability to generalize is the whole point of machine learning.
Challenge 2: Overfitting/High Variance	Low training error but high test error.	Regularization	The primary technique for controlling model complexity and reducing variance.
Challenge 3: Ill-posed Problems	Loss function is flat or parameters are highly correlated, making optimization difficult.	Ridge Regularization (as it makes the Hessian invertible/stable).	A practical reason for using regularization beyond just generalization.

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3. Regularization: Ridge (L_2) and Lasso (L_1)

Regularization adds a penalty term to the loss function to constrain the magnitude of the parameters $\mathbf{w}$.

- **General Regularized Objective:**

$$J(\mathbf{w}) = L(\mathbf{w}) + \lambda \Omega(\mathbf{w})$$

where $\Omega(\mathbf{w})$ is the regularizer and λ is the hyperparameter controlling the strength of regularization.

Regularizer	Penalty Term	Key Property
Ridge (L_2)	$\Omega(\mathbf{w}) = \frac{\lambda}{2} \ \mathbf{w}\ _2^2$	
LASSO (L_1)	$\Omega(\mathbf{w}) = \lambda \ \mathbf{w}\ _1$	

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The three challenges mentioned here did not follow the sequence as outlined in my notes. Instead, they independently summarized three challenges, and it might give the impression that each challenge has exactly one solution, while in reality the same technique (such as regularization) often addresses multiple issues. However, in reality, the thought process emphasized in EECS182 has been omitted.

The first long paragraph is very dense, with many concepts separated by vertical bars |. The structure is helpful for a table, but as prose it can feel overwhelming

4. Gradient Descent (GD) and Optimization

Gradient Descent is the core iterative algorithm used to find the minimum of the loss function $L(\mathbf{w})$. | **Concept** | **Formula/Explanation** | **Exam-Critical Rationale** | | :--- | :--- | :--- | | **GD Update Rule** | $\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla L(\mathbf{w}_t)$ | The **most fundamental optimization equation**. You must know how $\mathbf{w}$ is updated using the negative gradient and the learning rate η . | | **Learning Rate (η)** | A hyperparameter that controls the step size in the direction of the negative gradient. | Directly impacts **convergence speed** and **stability**. Must be tuned (or scheduled). | | **Linear Regression Gradient** | For $L(\mathbf{w}) = \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2$, the gradient is $\nabla L(\mathbf{w}) = 2\mathbf{X}^T(\mathbf{X}\mathbf{w} - \mathbf{y})$. | The specific gradient calculation for the most common example (**Least Squares**). | | **Full GD vs. Stochastic GD (SGD)** | **GD** uses the gradient of the entire dataset (**full-batch**). **SGD** (or **Mini-Batch GD**) uses the gradient of a small random subset of data (batch size B) to approximate the full gradient. | **SGD** is the **practical** method for large datasets due to computational cost, though it introduces more noise. |

Convergence of Gradient Descent

The convergence rate is influenced by the **eigenvalues** of the data matrix ($\mathbf{X}^T\mathbf{X}$) or Hessian.

- A key stability condition is that the learning rate η must be sufficiently small:
 $\eta \lambda_{\max}(\mathbf{X}^T\mathbf{X}) < 1$.
- $\eta^* = \frac{1}{\lambda_{\min} + \lambda_{\max}}$ is related to the optimal convergence rate for specific conditions. This demonstrates the link between $\mathbf{X}$'s properties and the **convergence speed**.

5. Ridge Regression Solution Details

Ridge regression for linear models has a closed-form solution:

$$\mathbf{w}_\lambda = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$$

- **Singular Value Decomposition (SVD) Interpretation:**

$$\mathbf{w}_\lambda = \sum_{i=1}^n \frac{\sigma_i}{\sigma_i^2 + \lambda} (\mathbf{u}_i^T \mathbf{y}) \mathbf{v}_i$$

- $\mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^T$
- This formula shows that the regularization term λ **dampens** components corresponding to **small singular values** (σ_i).
- Small σ_i 's are often associated with directions of low variance/high noise, which can lead to large weights in the unregularized solution ($\lambda = 0$)—this is the **large variance** issue that Ridge corrects.

In my notes, I have detailedly written out the derivation process here. From my personal perspective, I am not good at deriving formulas. However, in this case, Gemini did not follow the prompt's instructions and retain my personal preferences.

⚠ Content Emphasis Assessment

Emphasized Content in Notes	Likely Assessment Centrality	Rationale
SVD expansion of the Ridge solution ($\mathbf{w}_\lambda = \sum_{i=1}^n \dots$)	HIGH	This formula is fundamental to understanding how Ridge regularization works mathematically (by shrinking components corresponding to small singular values/directions of low data variance).
η^* formula ($\eta^* = \frac{1}{\lambda_{\min} + \lambda_{\max}}$)	MEDIUM	Demonstrates deep theoretical knowledge of GD convergence but is often considered secondary to the basic GD update rule and understanding the effect of the learning rate.
"Implicit Regularization" and using SVD in the GD update rule.	LOW-MEDIUM	This is a highly advanced concept linking GD to regularization and may be covered to show a theoretical connection (GD's "early stopping" acts as regularization), but it's less central than the explicit regularization formulas and the basic GD update. Flagged for less exam-centrality.

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4. Evaluation: Does This Replace Traditional Review?

Strengths.

Gemini goes beyond simple transcription or summarization and attempts to reconstruct structure, prioritize concepts, and explicitly reason about exam relevance, which aligns well with the intended use as an AI-enhanced review tool.

Limitations.

1. Gemini's notion is **largely inferred from mathematical density**, explicit formulas, and canonical ML structure, rather than from instructor-specific emphasis. As a result, it sometimes overestimates the centrality of theoretically deep but pedagogically secondary topics.
2. When I was taking the machine learning course, I used ChatGPT 4 to help me understand the formulas. I noticed that ChatGPT had this problem at that time, and today it has reappeared in Gemini. Our sources of images mainly included textbooks, course materials, and notes. For the same mathematical problem, the agreed-upon symbols during the derivation process were inconsistent. **I found that current large language models perform poorly in balancing the knowledge in the database and the content input by users.** Specifically, they tend to mix up these symbols or even completely ignore the symbols used by the users. This has caused great difficulties for beginners using AI tools to understand code.

Below are the prompts and corresponding conversations that I proposed to alleviate the aforementioned symbol inconsistency issue:

Do NOT re-derive the formula.
Instead, explain each term in the formula using my notation,
line by line, in plain language.

- ✦ That's a great approach to focus on understanding the components. I will explain the key formulas from your notes using your existing notation.

🧠 Explanation of Key Formulas

1. Ridge Regression Closed-Form Solution

This formula gives the optimal weight vector $\mathbf{w}_\lambda$ that minimizes the regularized squared-loss objective:

$$\mathbf{w}_\lambda = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

- $\mathbf{w}_\lambda$: The final vector of **model weights** (parameters) that the Ridge Regression algorithm has learned. The subscript λ denotes that the solution depends on the regularization strength.
 - $\mathbf{X}^T$: The **transpose of the data matrix $\mathbf{X}$** . If $\mathbf{X}$ has n samples and d features, $\mathbf{X}^T$ is a $d \times n$ matrix.
 - $\mathbf{X}$: The **data matrix** (or design matrix), where each row is a training example $\mathbf{x}_i$, and each column is a feature.
 - $(\mathbf{X}^T \mathbf{X})$: The **covariance matrix** (or Gram matrix) of the features (scaled by n). This $d \times d$ matrix is central to the Least Squares solution; its inverse is needed to solve the system.
 - $+$: Standard matrix addition.
 - λ : The **regularization strength hyperparameter** (the Greek letter lambda). It controls how much the model penalizes large weights. A larger λ means more aggressive weight shrinkage.
 - $\mathbf{I}$: The **Identity Matrix**. It is a $d \times d$ matrix with ones on the main diagonal and zeros elsewhere, ensuring that λ can be added to the diagonal of $\mathbf{X}^T \mathbf{X}$.
 - $(\dots)^{-1}$: The **matrix inverse** operation. This is applied to the $\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I}$ term. Adding $\lambda \mathbf{I}$ guarantees this matrix is invertible and numerically stable, which is a major advantage of Ridge.
 - $\mathbf{y}$: The **vector of target values** (labels or outputs) from the training data.
 - $(\dots)^{-1} \mathbf{X}^T \mathbf{y}$: This multiplication yields the final weight vector. The term $\mathbf{X}^T \mathbf{y}$ represents the correlation between the features and the target variable.
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2. Least Squares (Linear Regression) Gradient (Image 1)

This is the gradient of the squared loss function $L(\mathbf{w}) = \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2$:

$$\nabla_{\mathbf{w}} L_{\mathbf{w}^3} = 2\mathbf{X}^T(\mathbf{X}\mathbf{w} - \mathbf{y})$$

- $\nabla_{\mathbf{w}} L_{\mathbf{w}^3}$: The **gradient vector** of the loss function with respect to the weights $\mathbf{w}$. It indicates the direction of steepest increase in the loss.
 - 2: A scalar factor resulting from the derivative of the squared term.
 - $\mathbf{X}^T$: The **transpose of the data matrix**.
 - $(\dots)$: Parentheses grouping the term corresponding to the residual error.
 - $\mathbf{X}\mathbf{w}$: The **model's predictions** for the training data based on the current weights $\mathbf{w}$.
 - $\mathbf{y}$: The **true target vector**.
 - $(\mathbf{X}\mathbf{w} - \mathbf{y})$: The **residual vector**, representing the difference between the predictions and the true targets.
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3. Gradient Descent (GD) Update Rule (Image 1)

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t, \mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}})$$

- θ_{t+1} : The **updated model parameter vector** (weights) after iteration t .
 - θ_t : The **current model parameter vector** at iteration t .
 - $-$: Subtraction, moving the parameters in the direction opposite to the gradient.
 - η : The **learning rate** (eta), a scalar defining the step size for the update.
 - $\nabla L(\dots)$: The **gradient vector** of the loss function L .
 - $\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}}$: The **training data inputs (X) and targets (y)**, which the loss function is calculated over.
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4. Stochastic Gradient Descent (SGD) Update Rule (Image 2)

$$\theta_{t+1} = \theta_t - \frac{\eta}{B} \sum_{i=1}^B \nabla_{\theta_t} L_i(\theta_t)$$

5. Ridge Regression GD Update (Simplified Form) (Image 2)

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla L_{\text{ridge}}(\mathbf{w}_t)$$

- $\mathbf{w}_{t+1}$ and $\mathbf{w}_t$: The **updated** and **current** weight vectors.
 - η : The **learning rate**.
 - $\nabla L_{\text{ridge}}(\mathbf{w}_t)$: The **gradient** of the **full Ridge Regression objective function** (loss + penalty) evaluated at $\mathbf{w}_t$. This includes both the data fitting term (Least Squares gradient) and the weight shrinking term ($2\lambda\mathbf{w}_t$).
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6. Ridge GD Update Detail (Weight Decay Term) (Image 2)

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta [2\mathbf{X}^T(\mathbf{X}\mathbf{w}_t - \mathbf{y}) + 2\lambda\mathbf{w}_t]$$

- $\mathbf{w}_{t+1}$ and $\mathbf{w}_t$: The **updated** and **current** weight vectors.
 - η : The **learning rate**.
 - $2\mathbf{X}^T(\mathbf{X}\mathbf{w}_t - \mathbf{y})$: The **gradient of the data loss** (Least Squares term), driving the weights to fit the data.
 - $2\lambda\mathbf{w}_t$: The **gradient of the L_2 regularization penalty** term. This term acts like **weight decay**, pulling the current weights $\mathbf{w}_t$ directly towards zero on every step.
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7. Optimal Convergence Rate Factor (Image 1)

$$\eta^* = \frac{1}{\lambda_{\min} + \lambda_{\max}}$$

- η^* : The **theoretically optimal learning rate** (eta star) for fastest convergence in certain quadratic minimization problems.
 - 1: The numerator (constant one).
 - $\lambda_{\min}$: The **smallest eigenvalue** of the $\mathbf{X}^T\mathbf{X}$ matrix (or the Hessian of the loss function).
 - $\lambda_{\max}$: The **largest eigenvalue** of the $\mathbf{X}^T\mathbf{X}$ matrix.
 - $\lambda_{\min} + \lambda_{\max}$: The sum of the smallest and largest eigenvalues, which directly determines the optimal step size.
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8. SVD Interpretation of Ridge Solution Term (Image 2)

$$\mathbf{w}_\lambda = \sum_{i=1}^n \frac{\sigma_i}{\sigma_i^2 + \lambda} (\mathbf{u}_i^T \mathbf{y}) \mathbf{v}_i$$

- $\mathbf{w}_\lambda$: The optimal weight vector.
- $\sum_{i=1}^n$: A **summation** over all n singular values/components of the data matrix $\mathbf{X}$.
- $\mathbf{v}_i$: The i -th **right singular vector** of $\mathbf{X}$ (a column in the $\mathbf{V}$ matrix), representing a direction in the feature space.
- $\mathbf{u}_i^T \mathbf{y}$: The projection of the target vector $\mathbf{y}$ onto the i -th **left singular vector** of $\mathbf{X}$ (a row in $\mathbf{U}^T$), representing how much of the target data aligns with the i -th component.
- σ_i : The i -th **singular value** of the data matrix $\mathbf{X}$. Larger σ_i means more variance/information is captured by this component.
- $\frac{\sigma_i}{\sigma_i^2 + \lambda}$: This is the **shrinkage factor** applied to the i -th component. This term is small when σ_i is small (low data variance), showing that Ridge heavily shrinks components with low data support, which prevents overfitting.

5. Executive Summary

Gemini effectively transforms handwritten lecture notes into structured and readable study materials. **I think the understanding level of handwritten notes by Gemini is not significantly lower compared to the understanding of screenshots from electronic course materials.** It successfully highlights core concepts but tends to over-weight ideas that are emphasized in the student's notes rather than correcting for possible omissions or biases. The tool is most useful for post-lecture consolidation and organization, helping learners reconstruct the conceptual structure of a topic. Hallucination risk is low at the factual content level but moderate when identifying conceptual importance and relationships.

However, I would like to point out that, especially for courses with a lot of mathematical derivations like deep learning, large models "complete the most likely language patterns", and **the ones that are activated first are the "standard derivation templates" most common in the training data**, resulting in inconsistent symbolic expressions, which is difficult for beginners. Therefore, I suggest using AI tools more for review, and we can also use some prompts to guide the model.